Bridging the Technological Access Gap:

The Critical Role of Communication Training in Healthcare Delivery

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Introduction

In modern healthcare, digital technology has significantly transformed the way patients receive medical services. Tools such as telehealth consultations, electronic health records, patient portals, and mobile health applications have progressed from experimental ideas to common practices (Gajarawala & Pelkowski, 2021, p. 221). Nonetheless, this digital shift presents a paradox: while technology offers the potential for enhanced healthcare delivery and broader accessibility, it concurrently creates barriers for individuals who lack digital literacy or reliable access to technology (Panahi & Falkner, 2025, p. 2). This inequality, often referred to as the digital divide, risks exacerbating existing health disparities among at-risk populations, such as low-income individuals, older adults, rural residents, and people with limited education who may already experience structural barriers to care (Raihan et al., 2025, p. 111).

The COVID-19 pandemic significantly sped up the embrace of digital health, bringing the digital divide to the forefront (Vudathaneni et al., 2024, p.1). As healthcare systems rapidly shifted to digital formats, countless patients discovered they were unable to access fundamental medical care due to technological hurdles such as unreliable internet access, lack of appropriate devices, limited digital literacy, language barriers, and accessibility limitations (Gajarawala & Pelkowski, 2021, p. 218).

This paper focuses on the research question: What is the role of communication training in helping healthcare providers address technological access gaps in healthcare delivery, and how can such training be optimized to support vulnerable populations? As digital health continues to expand, providers must be equipped not only with clinical expertise but also with the communication skills necessary to help patients navigate digital systems. The following sections present a review of relevant literature, followed by an analysis of training approaches and recommendations for improvement.

Theoretical Framework

This examination draws on Digital Divide Theory and Health Communication Theory to frame how technological inequities persist in modern healthcare. Digital Divide Theory acknowledges that disparities in access exist at multiple levels, not only in infrastructure, but also in digital skills, usage patterns, and outcomes (Raihan et al., 2025, p. 112). In other words, access alone is not enough. While expanding broadband or distributing devices is important, these measures do not automatically ensure meaningful participation in digital healthcare systems. Wang et al. argue that the "characteristics of rural populations, rather than lower technology penetration in rural areas, account for the differences" in internet usage (2011, p. 8). I find this distinction especially important because it shifts the conversation away from purely technical solutions and toward human-centered factors. These characteristics may include older age demographics, lower average income, limited educational opportunities, reduced digital literacy, and less routine exposure to technology in work or community settings. Together, these factors influence confidence, trust, and familiarity with digital platforms, suggesting that closing the divide requires more than infrastructure; it requires intentional communication and support.

Health Communication Theory further strengthens this framework by emphasizing that effective healthcare depends on how well providers understand and respond to patients' specific needs (Pagano, 2019, p. 45). However, I argue that this principle is often acknowledged in theory but underapplies in practice. From my perspective, this theory highlights that technology cannot function independently of human interaction. Patients' needs may include step-by-step guidance for navigating patient portals, simplified explanations of medical terminology, or reassurance about privacy and data security. Cultural contexts may shape how individuals perceive authority, healthcare institutions, or digital surveillance, which can influence trust and willingness to engage. Communication barriers can include limited English proficiency, low health literacy, sensory impairments, or anxiety about using unfamiliar technologies. As Moorhead et al. note, digital health tools "bring a new dimension to health care as it offers a medium to be used by the public, patients, and health professionals to communicate about health issues with the possibility of potentially improving health outcomes" (2013, p. 1). However, I argue that this potential can only be realized when healthcare providers are trained to bridge technological systems and patient understanding through adaptive, culturally responsive communication strategies.

Literature Review

The Digital Divide represents a complex web of interrelated disadvantages that extend beyond lack of internet access (Raihan et al., 2025, p. 115). These disadvantages often overlap, including low income, limited education, and low digital literacy, as they cannot be reduced to a single issue or solution. Individuals facing one barrier frequently experience several at once, creating compounded obstacles to digital healthcare participation.

Wang et al. report that "individuals with medical conditions were far less likely to report Internet use than those without medical conditions (32.6% vs 70.3%)" (2011, p. 3). Although this finding seems surprising since individuals with medical conditions might be expected to search for health information online, it may reflect underlying demographic factors. Those with chronic conditions are often older, have lower incomes due to medical expenses, or face physical and cognitive limitations that make digital navigation difficult. These are the patients who could benefit most from telehealth and online health tools (Vudathaneni et al., 2024, p. 2), highlighting a critical gap between need and access.

Geographic disparities further illustrate this pattern. While rural residents initially appeared less likely to use the Internet than urban residents (59.7% vs 69.4%) (Wang et al., 2011, p. 3), this difference disappeared when demographic factors were controlled (p. 5). This suggests that characteristics such as older age, lower income, and lower educational attainment, rather than rural location alone, shape digital access.

Digital Health

Despite obstacles, digital health provides significant advantages when adopted fairly. Moorhead et al. outlined six main benefits of digital health, including improved access and enhanced communication (2013, p. 9). While these advantages exist, I am cautious about how they are often presented as universally accessible outcomes. In my view, these benefits assume a level of digital competence that many patients do not possess. This creates a gap between the promise of digital health and the lived experiences of vulnerable populations, who may encounter these systems not as empowering tools but as additional barriers. Vudathaneni et al. showed that telehealth resulted in "notable enhancements in patient health, including a reduction in disease-specific markers, a considerable decrease in symptom severity, and an overall improvement in health status," along with financial savings (2024, p.1).

However, Moorhead et al. (2013, p. 10) identify twelve drawbacks of digital health tools, including concerns about information quality, reliability, privacy, user comprehension, and the potential for miscommunication. Among these, quality concerns and reliability are the most pressing. Gajarawala and Pelkowski (2021, p. 219) highlight that "in comparison to in-person visits, telemedicine appointments face greater risks related to privacy and security." Additionally, technical infrastructure poses challenges; Upadhyay et al. (2023, p. 1) note that concerns related to energy consumption (power absorption) and accuracy are significant concerns in the deployment of smart healthcare technologies, which can be especially problematic for older patients or those with limited technical knowledge.

Communication as Bridge

Healthcare providers must be reconceptualized as crucial intermediaries who can either facilitate or hinder patients' effective use of digital health tools (Raihan et al., 2025, p. 120). I strongly agree with this perspective, but I extend it further by arguing that providers are not just intermediaries; they are gatekeepers of access. In digital health environments, patients' ability to receive care often depends on how effectively providers communicate instructions, build confidence, and respond to confusion. Without this support, access to technology does not translate into access to care. Moorhead et al. emphasize that effective digital health communication requires "understanding of audience needs, cultural contexts, and communication barriers" (2013, p. 2), understanding that requires human judgment, cultural competence, and empathy. Communication training can equip providers with strategies for assessing patients' digital literacy supportively. Rather than assuming all patients can navigate portals or video consultations, trained providers can ask questions like "Have you used a video call before?" or "Would you like me to walk you through accessing your test results online?" Gajarawala and Pelkowski emphasize that successful telehealth requires providers to "practice telehealth etiquette" and "be aware of and practice" communication standards (2021, p. 219).

Real-World Consequence

Cases such as Lauren Lowrey, a news anchor for WSMV Channel 4 in Nashville, Tennessee, who developed severe preeclampsia complications after giving birth and Jasmine Turner, a mother whose experience with preeclampsia was shared, described how delays in diagnosis occurred in part because she did not know which symptoms were urgent and how to communicate them effectively to her care team (“Jasmine Turner”).

While these examples are not primarily about digital literacy or access, they *do* underscore that effective communication is central to safe healthcare. Digital health technologies have significant potential to strengthen communication through symptom-tracking apps, automated reminders, secure messaging, and remote monitoring, but these tools cannot improve outcomes unless patients know how to use them and providers know how to interpret and act on the information. When patients lack reliable internet access, cannot afford devices, or are unfamiliar with digital tools, they risk being excluded from this digital layer of care, widening existing service gaps. This exclusion disproportionately affects the elderly, rural residents, low-income individuals, racial and ethnic minorities, and immigrants (Vudathaneni et al., 2024, p. 3). For digital health solutions to fulfill their promise, communication systems, both digital and human, must be designed to support patients’ understanding, confidence, and access.

Communication Training as Intervention

Although communication is vital, training for healthcare providers is still insufficient. Conventional medical and nursing education primarily emphasizes clinical knowledge and technical skills, often relegating communication training to a lesser priority (Pagano, 2019, p. 34). When communication training is provided, it generally concentrates on face-to-face interactions rather than the challenges posed by digital health.

In a discussion for this study, nursing student Busola Olope highlighted this deficiency. When I interviewed her about the communication training in her curriculum, Olope stated: "This semester, we had one course called mental health, where we were instructed in therapeutic communication. We learned how to build rapport with our patients and so on" (Personal interview). Nonetheless, she promptly recognized limitations, remarking: "What you're learning in class is different from encountering the real world." She stressed that communication is "something you engage in daily without truly getting accustomed to it. It's not merely being in a classroom and being expected to, you know, understand everything" (Personal interview).

"The clinicals we had this semester were quite different from how we applied what we learned in class with an actual patient" (Personal interview). What stood out to me in Olope's response is the clear divide between theoretical training and real-world application. While communication is taught in structured settings, I feel like training remains insufficient because it does not fully prepare providers for the unpredictability of patient interactions, especially in digital environments. Her experience reinforces my claim that communication is not a static skill but one that must be practiced and adapted continuously. While nursing programs are integrating communication training, the focus continues to lean toward traditional in-person encounters rather than digital situations, indicating that students require more realistic hands-on practice using telehealth platforms, patient portals, and other digital systems to develop the skills necessary for effective communication in virtual healthcare settings.

Moorhead et al. note that "health professionals may not frequently utilize social media to communicate with their patients" (2013, p.8), pointing to substantial gaps. Providers may be untrained, worried about legal or privacy issues, or simply lack the time to incorporate digital communication into an already demanding workload. The pandemic accelerated telehealth adoption without sufficient preparation, exposing critical shortcomings in providers' capabilities to communicate effectively via digital platforms (Gajarawala & Pelkowski, 2021, p. 219).

Elements of Effective Training

Effective training should encompass various areas. Healthcare providers require a sufficient level of technical literacy to assist patients in using basic technology, recognize common obstacles, offer initial troubleshooting, and know when to seek technical support. Panahi and Falkner emphasize that providers need "adequate training to use telemedicine platforms and interpret AI-generated insights effectively" (2). In terms of the importance of communication training for healthcare professionals, Olope highlighted its essential nature, sharing that her training emphasized the key principle of building rapport with patients before any clinical interaction (Personal interview). She noted that without structured communication education, she might have approached patient care solely from a technical perspective focused on treatment. This perspective highlights how formal communication training can fundamentally change providers' approach from a strictly technical model to one centered on relationships, recognizing the therapeutic benefits of the provider-patient connection.

In technical and professional communication research, empathy is described as the ability to understand another person’s perspective and respond in ways that acknowledge their needs and context (Petersen, 2017). Importantly, scholars note that empathy can be learned and strengthened through practice and training rather than being purely innate. In healthcare education, more specifically, evidence shows that training in empathy and patient-centered communication improves clinicians’ relational skills and supports more effective interactions with patients.

Training should also focus on sustaining therapeutic relationships in digital environments. Wang et al. reveal that among internet users with health issues, "more than two-thirds (68.3%) reported accessing the Internet only from home" (2011, p. 5), indicating that many patients interact with digital health resources in private settings where they may need immediate assistance.

Training must incorporate health literacy alongside digital literacy. Moorhead et al. indicate that "the general public may not know how to apply information found online to their personal health situation correctly" (2013, p. 10). Olope also acknowledged this, stating, "miscommunication can cause a lot of things in the healthcare setting" (Personal interview). She stressed that "healthcare workers should strive to communicate as clearly and compassionately as possible," recognizing that "sometimes it depends on the patient because some individuals are dealing with significant challenges that may lead them to vent their frustrations on others" (Personal interview).

Evidence-Based Strategies

The "teach-back" method can be adapted for digital contexts. After explaining portal access, providers ask patients to describe the steps they will take, identifying misunderstandings before independent home use. Plain language principles are essential in situations where patients cannot ask immediate questions. As Russell Willerton explains, “through dialogic ethics…the importance of clarity becomes paramount,” especially in BUROC (Bureaucratic, Unfamiliar, Rights-oriented and Critical) contexts where users cannot seek clarification (2015, p. 92). Gajarawala and Pelkowski emphasize that effective telehealth requires “clear standards for the acceptable format quality for medical images and data” that patients understand (2021, p. 219).

Cultural competence is crucial. Wang et al.’s (2011, p. 5) finding that racial disparities persisted after controlling for socioeconomic factors suggests that cultural factors such as patient beliefs, language preferences, trust in healthcare systems, and experiences of discrimination complexly influence technology adoption. Training should address how these cultural factors shape attitudes toward digital health, how language barriers compound technological barriers, and how prior experiences of discrimination may affect willingness to adopt digital technologies.

Olope recognized communication's importance in closing patient-provider gaps. When asked whether communication training could reduce barriers, she responded: "That's a big one, I would say a very big yes to that." She emphasized, "communication between nurses to nurses could also help the patients, not just the patients talking to the nurse, communication is a very big deal" (Personal interview), highlighting how training benefits both patient interactions and care coordination.

Implementing Communication Training

Consider a regional healthcare system serving diverse urban and rural populations. Regional Health Partners recognized that its rapid digital health adoption had created significant barriers for elderly patients, rural residents, non-English speakers, low-income patients, and patients with disabilities. Patient satisfaction scores declined, portal adoption remained low, and staff reported frequent frustrated patient calls.

The system developed comprehensive communication training for all clinical staff. First, technical orientation to organizational digital tools ensured staff could demonstrate functions, explain features, troubleshoot problems, and guide patients. Staff used all tools from patient perspectives, understanding patient experiences, and identifying confusing elements.

Second, extensive role-playing exercises are practiced, assessing technological capabilities and adapting communication. Scenarios included diverse patient populations with varying technological needs. To structure these efforts, the Practical Intervention Model offers a framework for addressing barriers systematically. The model emphasizes assessing patient needs, tailoring interventions to specific populations, providing ongoing support, and evaluating outcomes to refine strategies. By applying this model, Regional Health Partners ensured training was not only comprehensive but also adaptable to the needs of diverse patients and staff, creating a more targeted and effective intervention (Doueiri, Bajra, Srinivasan, Schillinger, & Cuan, 2024).

Third, training emphasized cultural humility and health equity, showing that assumptions about patients' technological abilities often reflect provider biases rather than actual skills. Cultural humility involves self-reflection, recognizing power imbalances, and understanding patients' perspectives. Providers are encouraged to ask questions and adapt support to each patient. Raihan et al. note that vulnerable populations, including older adults, low-income individuals, rural residents, and racial and ethnic minorities, face unique barriers to digital health access (2025, pp. 115–120). Integrating cultural humility helps providers address these challenges and build patient trust.

Outcomes and Challenges

Vudathaneni et al. (2024) report that communication-centered training led to measurable improvements. In their study, patient portal activation increased by 35% over six months, with significant gains among elderly patients (25% to 45%), limited English proficiency patients (18% to 38%), and low-income patients (30% to 52%). Telehealth satisfaction improved from 3.2 to 4.5 on 5-point scales.

However, challenges emerged. Some older physicians resisted serving as technology facilitators, viewing it as outside their role. Time constraints added pressure to clinical encounters. The system responded by extending appointments for patients needing technology support, creating group training sessions, developing video tutorials, and establishing technology support phone lines. Investment was justified by reduced missed appointments (25% decrease), emergency visits (18% decrease), and readmissions (15% decrease) (Vudathaneni et al., 2024, p. 5).

Upadhyay et al. (2023) observe that implementing healthcare technologies requires addressing "computational duration, accuracy and development issues" (p. 2). Communication training alone cannot overcome systemic barriers; it must be supported by technical support services, appropriate time allocation, organizational commitment, and payment models compensating for additional time required.

Best Practices

Training must be ongoing as technologies and patient needs evolve. Regional Health Partners implemented quarterly refreshers addressing new technologies, emerging challenges, platform updates, and evolving practices. Panahi and Falkner (2025) emphasize that providers need continuous education as "technology continues to advance and healthcare systems evolve" (p. 3). These refreshers also provide opportunities to reinforce effective communication strategies, ensuring staff can guide patients through digital tools clearly and compassionately.

Training is most effective when role-specific, practice-based, and focused on communication strategies. Different staff face different challenges: physicians must communicate clinical decisions clearly through digital channels, and nurses provide education and coordinate care with patients. Practice-based learning through role-playing develops both communication and technical skills rather than abstract knowledge (Gajarawala & Pelkowski, 2021, p. 219).

Organizational support is essential. Staff need technical support resources, protected time for technology-oriented patient support, and leadership commitment prioritizing digital inclusion. Raihan et al. (2025) argue that addressing digital inequity requires "coordinated efforts at the federal, state, and local levels to guarantee that everyone has the required digital resources and access" (p. 123). By combining technical proficiency with communication-focused training, organizations can ensure that digital health adoption is both effective and equitable.

Communication Training in Context

While focusing on communication training, we must acknowledge that it alone cannot solve fundamental digital divide inequities. Wang et al.'s research demonstrates that internet access disparities reflect deep-rooted social, economic, and geographic inequality (2011, pp. 3-5). Addressing systemic issues requires policy interventions: expanded broadband infrastructure, subsidized internet programs, device distribution, and digital literacy investment.

However, communication training remains crucial. Even with improved infrastructure, the provider-patient interface remains critical, where digital access can be facilitated or hindered. Moorhead et al. observe that realizing digital health potential "requires healthcare providers who can navigate the intersection of technology and communication effectively" (2013, p. 8). Communication training represents an immediate intervention while advocating for broader systemic changes.

Addressing Persistent Disparities

Research by Wang et al. indicates that racial inequalities persisted even after accounting for socioeconomic factors (2011, p. 5), showing that barriers go beyond income or education. Cultural beliefs, such as preferring in-person guidance over apps, can affect digital engagement. Experiences of discrimination may lead to mistrust of healthcare systems, reducing telehealth use. Language obstacles arise when portals and instructions are only in English. A lack of culturally appropriate content occurs when online health information assumes Western health practices and diet advice that conflict with cultural food traditions, thereby ignoring diverse traditions or concerns. Training must address these issues rather than assuming that providing technology and basic instruction is enough.

For populations with limited English proficiency, language barriers significantly exacerbate technological challenges. Service providers need to receive training on how to work effectively with interpreters in digital settings, advocate for high-quality multilingual resources, and recognize when language barriers render digital tools unsuitable.

Upadhyay et al. indicate that technological infrastructure may systematically marginalize certain demographics (2023, pp. 2-4). Their observation that "sensors must be recalibrated regularly to ensure that they are functioning correctly" underscores the assumption in digital health that users possess adequate technical knowledge. Providers require training to identify when technologies may not be suitable for specific patients or situations.

Policy Implications

Communication training addressing digital health should be integrated into standard healthcare professional education. Medical schools, nursing programs, and other training programs should incorporate digital health communication modules throughout their curricula.
Professional licensing bodies could incorporate digital health communication competencies into licensure examinations. Because these competencies rely on clarity, audience awareness, and effective information design, healthcare education programs and institutions would benefit from involving professional and technical communicators in developing these modules. Their expertise supports the creation of user-centered training materials that help providers navigate digital platforms, explain technology to patients, and communicate effectively across virtual channels.

Healthcare organizations should include digital health communication competencies in performance evaluation and continuing education requirements. Performance criteria should include objective measures of provider support for digital health adoption alongside subjective patient assessments.

Government programs such as Medicare and Medicaid could incentivize training through payment models rewarding patient portal activation, successful telehealth encounters, and effective digital health engagement. Current payment models often discourage time investment in technology support. Gajarawala and Pelkowski note "the Bipartisan Budget Act was approved by Congress" to address "major changes for Medicare telehealth policy" (2021, p. 220).

Efforts expanding digital health access should be evaluated for health equity impacts, with requirements for assessing impacts on vulnerable populations. Upadhyay et al. note technological innovations risk widening disparities if not carefully implemented (2023, pp. 1–2). Raihan et al. emphasize that "prioritizing state investment in cost-effective broadband connection, targeted programs and training and culturally sensitive content" is essential (2025, p. 123).

Future Directions

Thorough evaluation studies are essential to assess the effectiveness of various training methods. Research should explore how training outcomes differ among patient demographics and healthcare settings. Longitudinal studies should focus on identifying organizational elements that facilitate lasting behavior change, as well as on interdisciplinary collaboration between PTC scholars and health systems. Additionally, cost-effectiveness research would aid organizations in prioritizing their investments. Vudathaneni et al. offer a model showing that telehealth led to "substantial reduction in direct and indirect healthcare costs" while also improving outcomes (2024, p. 5).

Conclusion

Technologies can increase accessibility, boost engagement, deliver essential information, and promote better health outcomes. Nevertheless, to achieve these benefits, it's crucial to ensure that tools support rather than marginalize vulnerable groups. This research demonstrates that communication training is not optional but essential for achieving equitable healthcare access. I argue that without intentional investment in communication training, digital health will continue to reproduce existing inequities rather than resolve them. While technological innovation is often framed as progress, my analysis shows that progress without accessibility is incomplete. Communication training, when combined with systemic reform, offers a practical and immediate way to bridge this gap.

Communication training tackles these issues by equipping providers with the knowledge, skills, and attitudes necessary to effectively facilitate digital health access. As nursing student Busola Olope pointed out, communication training is crucial for enabling providers to "build this rapport with your patients" and convey information "in the simplest and most caring way possible" (Personal interview). Her perspective underscores the increasing acknowledgment of the significance of communication and the persistent need for practical, clinically-focused training that prepares providers to meet patients' varied technological needs.

The theoretical framework that combines Digital Divide Theory and Health Communication Theory sheds light on why communication training, while necessary, is not enough for achieving digital health equity. The digital divide signifies deep-rooted systemic inequalities requiring policy measures: improved infrastructure, digital literacy initiatives, financial assistance, culturally relevant content, and anti-discrimination policies. Nonetheless, even with fundamental changes in the system, the quality of communication between providers and patients will continue to be a determining factor in whether digital health meets its potential.

Healthcare must prioritize inclusive digital health rather than focusing solely on technological advancement. This necessitates that providers recognize their responsibilities extend beyond providing clinical care to facilitating meaningful engagement with digital tools, organizations offering essential training and support, and policymakers acknowledging digital health access as a key factor in health equity. Communication training serves as one avenue toward ensuring that digital transformation enhances rather than detracts from healthcare's fundamental mission: addressing the health needs of all individuals regardless of their technological resources, digital literacy, socioeconomic status, geographic location, language, or cultural background. Therefore, communication training, when integrated with systemic reform, becomes a critical mechanism through which healthcare providers can actively reduce technological access gaps.

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